Computer Vision Winter 2026 - Exercise 4

Machine Learning for Computer Vision - TU Dresden

November 16 – 20, 2026

Logistic Regression

Recall the definition of the l_2-regularized logistic regression problem for linear functions $f_\theta(x_s) = \langle x_s, \theta \rangle$:

$$\min_{\theta \in \Theta} \varphi(\theta)$$

With the objective function $\varphi(\theta)$ defined as follows:

$$\varphi(\theta) = \lambda R(\theta) + \sum_{s \in S} L(f_\theta(x_s), y_s)$$

And L, R and λ as shown in the lecture.

$$\begin{aligned} L(f_\theta(x_s), y_s) &= -y_s f_\theta(x_s) + \log(1 + 2^{f_\theta(x_s)}) \\ R(\theta) &= \|\theta\|_2^2 \\ \lambda &= \frac{\log e}{2\sigma^2} \end{aligned}$$

In this exercise, we aim to learn a linear function f_θ that produces binary labels for pixels s in an image using logistic regression. For each pixel s, we are given a feature vector x_s, which contains the color values of all pixels in a neighborhood around s.
The first step in implementing logistic regression is to be able to calculate the linear function $f_\theta(x_s) = \langle x_s, \theta \rangle$.

1. Complete the implementation of the function `calculate_linear_function_output`.

Next we want to be able to calculate the current value of the objective function $\varphi(\theta)$

2. Complete the implementation of the function `calculate_objective_function`.

We will find a solution $\hat{\theta}$ to the l_2-regularized logistic regression problem by means of a steepest descent algorithm:

$\hat{\theta} := 0$
repeat
  $d := \nabla\varphi(\hat{\theta})$
  $\hat{\theta} = \hat{\theta} - \eta d$
  if $\|d\| < \epsilon$
    return $\hat{\theta}$

With some parameter η, usually called the learning rate and the stopping criterion parameterized by ϵ.

3. To implement the steepest descent algorithm we require the gradient $\nabla\varphi(\theta)$ of the objective function $\varphi(\theta)$. The gradient contains the partial derivatives $\frac{\partial\varphi}{\partial\theta_i}$ of $\varphi(\theta)$ with respect to each parameter θ_i. Analytically derive $\frac{\partial\varphi}{\partial\theta_i}$.

4. Using your result from task 3 for $\frac{\partial\varphi}{\partial\theta_i}$ complete the implementation of the function `compute_grad`.

5. Using the functions implemented in tasks 1-4 complete the implementation of the steepest descent algorithm. Test your implementation with the pixels of `flower.png` as samples and `flower_labels.png` as labels.

6. Analytically show that the objective function $\varphi(\theta)$ is convex.

Recall the defintion of the inference problem.

$$\min_{y \in \{0,1\}^S} \sum_{s \in S} L(f_\theta(x_s), y'_s)$$

7. For the input image `flower_test.png` and the parameters $\hat{\theta}$ learned in task 5 find the optimal solution to the inference problem. Separately display the pixels that have been assigned $y_s = 0$ and $y_s = 1$ respectively.